

Software Requirements Specification (SRS):

- **Introduction:**

- **Purpose:**

This SRS document clearly states the requirements for driving profitable business decisions, including revenue growth, optimizing operations, enhancing customer experience, and supporting decision-making across the organization.

- **Document Conventions:**

- **Bold** text is used for key business terms and section headings.
 - Requirements are labeled as **FR** (Functional Requirement) and **NFR** (Non-Functional Requirement).
 - The document follows IEEE standards for Software Requirements Specifications.

- **Intended Audience and Reading Suggestions:**

This document is catered towards the following teams in the organization:

1. **Stakeholders & Leadership** – To understand business value and ROI.
2. **Marketing Team** – For customer segmentation and campaign planning.
3. **Finance Team** – For pricing analysis and revenue optimization.
4. **Operations / Restaurant Partners** – For performance benchmarking.
5. **Data & Development Teams** – For implementation guidance.

- **Project Scope:**

- **In Scope:**

- Ingestion, Cleaning, and Feature Engineering on the provided dataset.
 - Generation of business insights on pricing, customer preferences, location performance, and ratings
 - Suggesting specific business recommendations to help the business make better decisions.
 - Support for cross-functional teams to improve profitability and competitiveness
 - Delivery of a scalable solution (website, mobile app, dashboard, ML models, or hybrid)

- **Out of Scope:**

- Using Machine Learning for performing predictive analytics.

- **References:**

- IEEE Std 830-1998 – Recommended Practice for Software Requirements Specifications.
 - Zomato Bangalore Restaurant Dataset (provided)

- **Overall Description:**

- **Product Perspective:**

The sole aim of this document is to act as a guide for teams, helping them better understand the business and its needs, and enabling them to derive meaningful solutions from large volumes of restaurant data. By preparing the data for analysis, the system will help the company identify high-performing restaurants, optimize pricing strategies, improve marketing effectiveness, and increase overall platform revenue and partner satisfaction.

This document will empower every team to make data-driven decisions that will increase platform revenue, partner retention, customer engagement, and market share in the food service industry.
- **Product Features:**
 - **Personalized Restaurant Discovery:** Recommending restaurants to users tailored to their dining preferences.
 - **Affordability:** Performing data analysis to help customers find good-quality restaurants at an affordable rate.
 - **Customer Behavior Trends:** Understanding popular cuisines, online ordering patterns, and booking preferences to improve user experience.
 - **Restaurant Performance Intelligence:** Developing tools that help restaurant partners understand customer feedback, their strengths and weaknesses, and areas of improvement.
- **User Classes and Characteristics:**
 - **End Customers:** Need to easily discover high-quality, affordable restaurants, along with personalized suggestions.
 - **Restaurants Partners / Owners:** Need performance insights, customer feedback analysis, and improvement recommendations.
 - **Work Teams:** Support customer-centric initiatives and business growth.
- **Operating Environment:**

The platform should be easily accessible via the website and mobile apps for maximum customer and partner reach.
- **Design and Implementation Constraints:**
 - The platform must be accessible via:
 - Website
 - Mobile applications (Android and iOS)
 - The platform shall support:
 - Cross-platform compatibility (mobile, tablet, desktop)
 - Internet-based access with reliable connectivity
 - 24×7 availability for all users
 - The system shall include structured data handling and analytics capabilities, including:
 - Data pipeline for data collection, cleaning, and transformation

- Exploratory Data Analysis for presenting insights.
 - Having a Dashboard Facility for clear data visualization and reporting.
 - Deploying Machine Learning models on the AWS cloud format (*as specified by the stakeholders).
- The platform must support authentication and role-based authorization for the following types of users:
 - Restaurant Owner
 - Restaurant Staff
 - Delivery Partner
 - Tech and Business Teams (Admin Users)
 - End Customers
- **Assumptions and Dependencies:**
 - **Assumptions:**
 - Users (customers, restaurant partners, and staff) will have basic digital literacy to interact with web and mobile applications.
 - Internet connectivity will be available for accessing the platform.
 - Any new data will get automatically updated with the help of data pipelines, which can then be used for both exploratory and predictive analytics.
 - Machine Learning models will improve over time as more data becomes available.
 - **Dependencies:**
 - Availability of a reliable and relevant dataset for analysis.
 - Cloud infrastructure services (e.g., AWS) for deployment, storage, and scalability of ML models.
 - Python programming environment for data processing, exploratory analysis, and machine learning.
 - Apache Spark for building and managing scalable data pipelines.
 - Flutter framework by Google for developing mobile apps.
 - React (frontend) and Django (backend) frameworks for web application development.
- **System Features:**
 - **Functional Requirements:**
 1. **User Management:**
 - a. User Authentication:
 - b. Role-based authorization.

Fields to include in the user-table:

 - a. User-ID:
 - i. Primary Key
 - ii. 13-digit long
 - iii. Auto-generated
 - b. User Name
 - i. 30 characters long
 - c. Age
 - d. Gender
 - e. DOB
 - f. Location
 - g. Email-ID
 - h. Phone-Number

2. Restaurant Discovery:

- a. Search Restaurants
- b. Filter Restaurants by:
 - i. Name
 - ii. Location
 - iii. Customer-rating
 - iv. Distance from the customer's residence
 - v. Food type
- c. View menus, ratings, and reviews

3. Order Management:

- a. **CRUD** (Create, Read, Update, Delete) Orders.
- b. Payment processing
- c. Order tracking

4. Delivery Management:

- a. Assign orders to delivery partners
- b. Real-time tracking

5. Restaurant Dashboard:

- a. Manage menu and orders
- b. View analytics

6. Recommendation System:

- a. Provide personalized restaurant suggestions.
- b. Suggestions must be based on:
 - i. Past order history
 - ii. User preferences (cuisine, cost range, location)
 - iii. Ratings and reviews given by the user
- c. The following ML techniques can be used for recommendation:
 - i. Collaborative Filtering
 - ii. Content-Based Filtering
 - iii. Hybrid models
- d. The system shall update recommendations dynamically when:
 - i. User preferences change
 - ii. New data is added to the system
 - iii. Restaurant ratings or availability changes
- e. The recommendations must be provided in the available timeslot of 5 seconds.

7. Reporting and Dashboards:

- a. The system shall provide comprehensive reporting and dashboard functionalities to support data-driven decision-making for business teams, restaurant partners, and internal stakeholders.
- b. The following types of Dashboards have to be made:
 - i. **Business Analytics Dashboard:**
The system shall provide dashboards for internal business teams displaying:
 - Total orders, revenue, and growth trends
 - Customer acquisition and retention metrics
 - Popular restaurants, cuisines, and locations
 - Peak order times and seasonal trends
 - ii. **Restaurant Performance Dashboard:**
The system shall provide restaurant partners with performance insights, including:

- Total orders and revenue generated
- Customer ratings and feedback analysis
- Popular menu items
- Order completion rates and cancellations

iii. **Delivery Performance Dashboard:**

The system shall provide insights for delivery operations, including:

- Average delivery time
- Order assignment efficiency
- Delivery partner performance metrics

- c. **(optional)** The system shall integrate Machine Learning models within the dashboard to provide users with easy access to real-time and data-driven recommendations.

• **External Interface Requirements:**

- **User Interfaces:**
 - Customer - friendly mobile and web interfaces.
 - Partner-friendly dashboards for restaurant owners.
- **Hardware Interfaces:**
 - *(not applicable)*
- **Software Interfaces:**
 - Zomato Android Mobile App
 - Zomato iOS Mobile App
 - Zomato Website
- **Communications Interfaces:**
 - 24/7 Customer Support is integrated into the platform with the OpenAI API.

• **Non-functional Requirements:**

- **Performance Requirements:**
 - The system shall respond to user requests within **2–3 seconds** under normal load conditions.
 - Real-time features such as **order tracking and recommendations** shall update with minimal latency.
 - Data processing pipelines shall complete batch processing within defined time limits.
- **Safety Requirements:**
 - High accuracy of insights to build customer trust.
 - Machine Learning models shall be periodically evaluated and updated to avoid incorrect or misleading outputs.
 - Error handling mechanisms shall ensure that failures do not result in incorrect information being shown to users.
- **Security Requirements:**
 - Sensitive data (passwords, personal details) shall be **encrypted and securely stored**.
 - The system shall implement **authentication and role-based authorization** for all users.
 - Secure payment processing shall be ensured through trusted payment gateways.
- **Software Quality Attributes:**

- **Customer Focus:** All features prioritize diner satisfaction and partner success.
- **Usability:** Intuitive for both end customers and business users.
- **Scalability:** The system shall support a **high number of concurrent users** without performance degradation.
- **Reliability:** Consistent and trustworthy insights.
- **Efficiency:** The system must use the device's resources.

Documentation prepared by:

- **Name:** Om Satyawar Pathak
- **Date:** 25 May 2026
- **Contact:** omsatyawanpathakwebdevelopment@gmail.com (or) omsatyawanpathakgit@gmail.com